Abu Bakar

AWS Certified Cloud Practitioner

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PROFESSIONAL SUMMARY

Backend software engineer with 2+ years of production experience designing distributed systems and cloud-based services. Currently a Research Assistant on a DFG-funded project while pursuing an MSc in Computer Science at Universität Rostock, with active side work in computer graphics and 3D computer vision. Actively pivoting into AI and machine learning through applied projects and graduate coursework. AWS Certified Cloud Practitioner.

WORK EXPERIENCE

Research Assistant (Wissenschaftliche Hilfskraft/HiWi)

Universität Rostock *November 2025 – Present*

Chair of Business Informatics (Prof. Michael Fellmann) — EDWANCE Project (DFG-funded)

- Architect and develop a full-stack distributed system for the DFG-funded EDWANCE research project, including a cross-platform .NET Desktop Activity Logging Tool that captures and structures user interaction data, validated through automated unit and integration test suites achieving >90% code coverage.
- Drive technical architecture and sprint-based task planning alongside a PhD researcher and supervising Professor, owning CI/CD pipeline quality with required pre-merge checks, clean .NET code, and technical documentation that bridge academic research requirements with production-grade software engineering standards.

Tech Stack/Tools: .NET, C#, Full-Stack, CI/CD, Unit & Integration Testing, Git, Agile/Sprint Planning, Cross-Platform Desktop Development

Software Engineer (Backend) - Remote

Careem Technologies (Uber, Inc and e& subsidiary) *December 2023 – April 2024*

- Developed and maintained backend services for the Amaken map platform in GoLang, leveraging PostgreSQL and PostGIS for high-performance spatial data modeling and geospatial querying.
- Delivered full-stack features end-to-end by building responsive React UI components and integrating them with internal RESTful APIs, validated through Swagger UI.
- Architected automated geographic-data workflows — including Overture dataset migration, GitHub Actions CI pipelines, AWS Route53, and Bifrost — and managed secure image storage and retrieval on AWS S3.
- Drove production reliability through a layered quality strategy — unit, integration, and acceptance tests achieving 80%+ code coverage (SonarQube) — plus end-to-end monitoring (Grafana, VictorOps, Argus UI) with Slack/on-call alerting.

Tech Stack/Tools: GoLang, PostgreSQL, PostGIS, React, AWS S3, GitHub Actions, Route53, Bifrost, Grafana, VictorOps, Argus UI, Swagger, SonarQube, JIRA, Kanban

Software Engineer (Backend) – Remote

NorthBay Solutions, LLC *September 2022 – November 2023*

- Developed and maintained web-parsing and Web API features in C#, .NET Core, and .NET MVC for the Auditor extraction module of Intelligize (a LexisNexis product), backed by Entity Framework data-access layers over SQL.
- Owned the AWS S3 document-processing pipeline end-to-end — extraction, parsing, data enrichment, and re-uploading of processed artefacts for frontend visualisation.
- Deployed and operated parsing-backlog microservices through a Jenkins-based CI/CD pipeline, validating releases via unit and integration tests and proactively resolving production defects to keep document throughput stable across batches.
- Collaborated in an Agile/Scrum environment — sprint planning, stand-ups, retrospectives, and code reviews — using JIRA for task tracking and Bitbucket for version control.

Tech Stack/Tools: C#, .NET Core, .NET MVC, Entity Framework, SQL, AWS S3, Bitbucket, JIRA, Agile, Jenkins

PROJECTS

VetraPath: Physically-Based Monte Carlo Path Tracer with Interactive Viewport *C++17, Dear ImGui, GLFW, OpenGL, Intel OIDN, CMake*

- **Rendering Engine & Materials:** Built a Monte Carlo path tracer from scratch with unbiased global illumination — Multiple Importance Sampling (MIS) with power heuristic, Next Event Estimation (NEE), and Russian Roulette termination — paired with a full BSDF library (Lambertian, rough metal, dielectric with Fresnel/refraction, GGX microfacet, emissive area lights) for physically-based rendering.
- **Spatial Acceleration:** Engineered a Bounding Volume Hierarchy (BVH) using the Surface Area Heuristic (SAH) over axis-aligned bounding boxes, reducing ray-scene intersection from O(n) to O(log n) on high-poly meshes.

- **Parallelism & AI Denoising:** Wrote a multi-threaded tile-based renderer using std::thread and integrated Intel Open Image Denoise (OIDN) to deliver production-quality, low-noise frames at significantly reduced sample counts.
- **Interactive Viewport & Asset Pipeline:** Designed a Dear ImGui + GLFW + OpenGL viewport with real-time progressive rendering, fly-camera, live material and lighting controls, and hot-reloading — backed by an OBJ/MTL mesh loader and tinyxml2-based scene serialisation for reproducible, version-controlled renders.

Tech Stack: C++17/20, CMake, Dear ImGui, GLFW, OpenGL, GLM, tinyxml2, Intel OIDN, stb_image.
Concepts: Ray Tracing, Monte Carlo Path Tracing, PBR, BVH/SAH, MIS, NEE, Multi-threading, Design Patterns.

RGB-D Semantic Scene Graph Generation with BIM/IFC Priors *Python, PyTorch, Grounding DINO, SAM 2.1, IfcOpenShell, NetworkX, Docker*

- **Open-Vocabulary Perception & 3D Fusion:** Built an end-to-end pipeline that transforms egocentric RGB-D sequences into 3D semantic scene graphs by pairing open-vocabulary 2D detection (Grounding DINO, Swin-Base) with stateful video mask propagation (SAM 2.1 Hiera-Large) across hundreds of frames, then back-projecting masked depth into world coordinates and deduplicating detections across views via centroid + bbox + voxel-IoU merging.
- **BIM/IFC Integration:** Extracted canonical structural entities (walls, doors, slabs, multi-storey hierarchies) from IFC building models with IfcOpenShell, and synthesised room polygons via point-cloud BEV (Bird's-Eye View) occupancy morphology when IfcSpace annotations were absent.
- **Scene Graph Construction:** Modelled each scene as a 4-layer hierarchical graph (Building → Storey → Room → Object) with spatial-relationship edges (proximity, alignment, above/below) built efficiently over a 3D KD-tree to avoid $O(N^2)$ lookups, plus door-to-wall portal edges extracted directly from the IFC topology.
- **Reproducibility & Visualisation:** Containerised the full pipeline with Docker + Make and shipped three rendering views per scene — 3D matplotlib + interactive Plotly HTML, top-down BEV projection, and a hierarchical tree — alongside GraphML and node-link JSON exports for downstream tooling (yEd, Gephi).

Tech Stack: Python, Make, PyTorch, Grounding DINO, SAM 2.1, IfcOpenShell, NetworkX, NumPy, SciPy, Open3D, Plotly, matplotlib.
Concepts: Open-vocabulary detection, multi-view geometry, point-cloud processing, BIM/IFC, semantic scene graphs, KD-tree spatial indexing, Docker.

SKILLS

Backend: GoLang, C#, .NET Core, .NET MVC, Entity Framework, RESTful APIs, SQL, PostgreSQL, PostGIS, Web Parsing, Microservices.

Frontend: React.js, HTML5, CSS3, JavaScript (ES6+), Swagger UI.

Graphics & Computer Vision: C++17, Modern OpenGL, Dear ImGui, GLFW, GLM, Monte Carlo Path Tracing, BVH/SAH, PBR/BSDF, Intel OIDN, CMake.

ML & 3D Perception: Python, PyTorch, Grounding DINO, SAM 2.1, IfcOpenShell, NetworkX, NumPy, SciPy, Open3D, Plotly, point-cloud processing.

Cloud & DevOps: AWS (S3, Route53), GitHub Actions, Bifrost, Jenkins, CI/CD, Docker, VictorOps, Grafana, Argus UI.

Testing & QA: Unit Testing, Integration Testing, Acceptance Testing, SonarQube, Swagger UI.

Tools & Workflow: Git, GitHub, Bitbucket, JIRA, Agile (Scrum & Kanban), Slack.

Other: Postman, Linux / Ubuntu, Google Firebase.

EDUCATION

Master of Science in Computer Science – Rostock, Germany
Universität Rostock *April 2024 – Present*

Bachelor of Science (Hons) in Computer Science – Lahore, Pakistan
Forman Christian College University (A Chartered University) *August 2018 – July 2022*

CERTIFICATIONS & LANGUAGES

Certification: AWS Certified Cloud Practitioner (Amazon Web Services).
Languages: English (C1 CEFR) | Deutsch (A2).